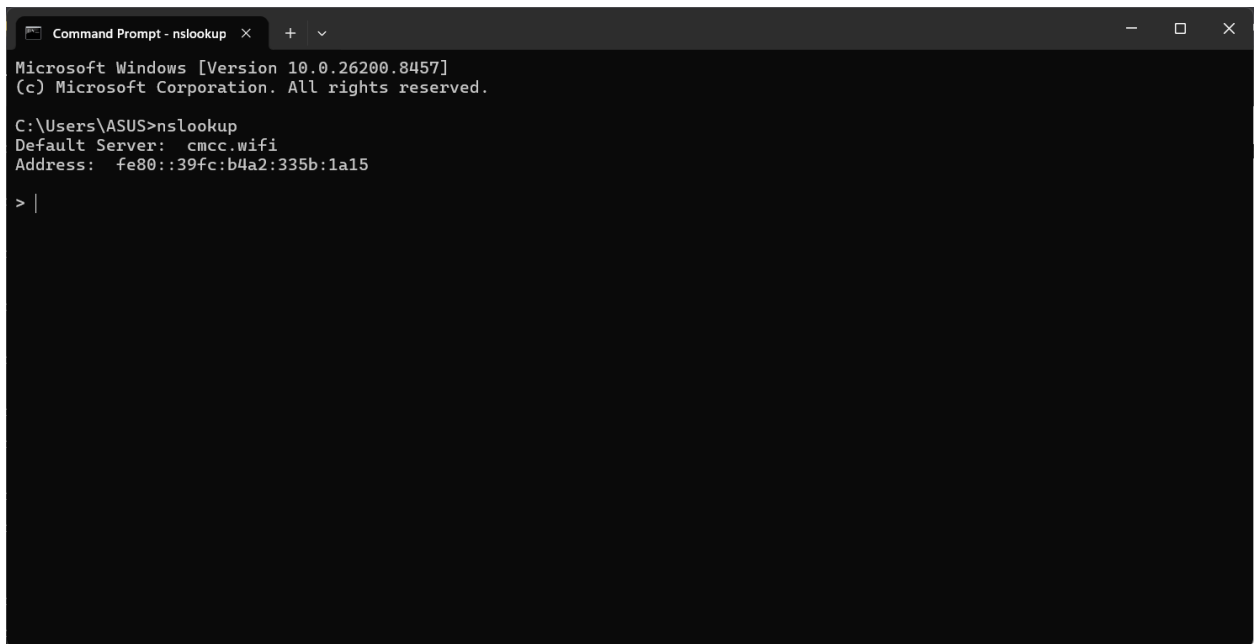


Chapter 17 Web Components, E-mail, and Instant Messaging

17.01 E-mail Headers

1a–1f

A

A screenshot of a Windows Command Prompt window titled "Command Prompt - nslookup". The window shows the output of the nslookup command. The text displayed is: "Microsoft Windows [Version 10.0.26200.8457] (c) Microsoft Corporation. All rights reserved. C:\Users\ASUS>nslookup Default Server: cmcc.wifi Address: fe80::39fc:b4a2:335b:1a15 > |".

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ASUS>nslookup
Default Server:  cmcc.wifi
Address:  fe80::39fc:b4a2:335b:1a15

> |
```

B

```
Command Prompt - nslookup  x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ASUS>nslookup
Default Server:  cmcc.wifi
Address:  fe80::39fc:b4a2:335b:1a15

> set q=mx
> |
```

C

```
Command Prompt - nslookup  x + v
Address:  fe80::39fc:b4a2:335b:1a15

> set q=mx
> rit.edu
Server:  cmcc.wifi
Address:  fe80::39fc:b4a2:335b:1a15

*** cmcc.wifi can't find rit.edu: Query refused
> > server 8.8.8.8
Unrecognized command: > server 8.8.8.8
> server 8.8.8.8
Default Server:  [8.8.8.8]
Address:  8.8.8.8

> set q=mx
> rit.edu
Server:  [8.8.8.8]
Address:  8.8.8.8

Non-authoritative answer:
rit.edu MX preference = 5, mail exchanger = mx03g-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03t-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03a-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03c-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03e-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03d-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03h-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03f-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03b-in01r.rit.edu
> |
```

I'm seeing 9 RIT SMTP servers

D

```
Command Prompt - nslookup  X + v
> set q=mx
> rit.edu
Server:  cmcc.wifi
Address:  fe80::39fc:b4a2:335b:1a15

*** cmcc.wifi can't find rit.edu: Query refused
> > server 8.8.8.8
Unrecognized command: > server 8.8.8.8
> server 8.8.8.8
Default Server:  [8.8.8.8]
Address:  8.8.8.8

> set q=mx
> rit.edu
Server:  [8.8.8.8]
Address:  8.8.8.8

Non-authoritative answer:
rit.edu MX preference = 5, mail exchanger = mx03g-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03t-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03a-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03c-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03e-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03d-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03h-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03f-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03b-in01r.rit.edu
> set q=a
> |
```

E

```
Command Prompt - nslookup  X + v
Unrecognized command: > server 8.8.8.8
> server 8.8.8.8
Default Server:  [8.8.8.8]
Address:  8.8.8.8

> set q=mx
> rit.edu
Server:  [8.8.8.8]
Address:  8.8.8.8

Non-authoritative answer:
rit.edu MX preference = 5, mail exchanger = mx03g-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03t-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03a-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03c-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03e-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03d-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03h-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03f-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03b-in01r.rit.edu
> set q=a
> mx03c-in01r.rit.edu.
Server:  [8.8.8.8]
Address:  8.8.8.8

Non-authoritative answer:
Name:  mx03c-in01r.rit.edu
Address:  129.21.10.162

> |
```

F

Finding the MX Records:

```
> set q=mx
```

```
> fingerlakes.edu
```

```
Command Prompt - nslookup x + v
Address: 8.8.8.8

Non-authoritative answer:
rit.edu MX preference = 5, mail exchanger = mx03g-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03t-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03a-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03c-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03e-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03d-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03h-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03f-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03b-in01r.rit.edu
> set q=a
> mx03c-in01r.rit.edu.
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
Name: mx03c-in01r.rit.edu
Address: 129.21.10.162

> set q=mx
> fingerlakes.edu
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
fingerlakes.edu MX preference = 10, mail exchanger = ms98607146.msv1.invalid.outlook.com
fingerlakes.edu MX preference = 0, mail exchanger = fingerlakes-edu.mail.protection.outlook.com
> |
```

Finding the A Record:

> set q=a

> fingerlakes.edu

```
Command Prompt - nslookup x + v
rit.edu MX preference = 5, mail exchanger = mx03f-in01r.rit.edu
rit.edu MX preference = 5, mail exchanger = mx03b-in01r.rit.edu
> set q=a
> mx03c-in01r.rit.edu.
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
Name: mx03c-in01r.rit.edu
Address: 129.21.10.162

> set q=mx
> fingerlakes.edu
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
fingerlakes.edu MX preference = 10, mail exchanger = ms98607146.msv1.invalid.outlook.com
fingerlakes.edu MX preference = 0, mail exchanger = fingerlakes-edu.mail.protection.outlook.com
> set q=a
> fingerlakes.edu
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
Name: fingerlakes.edu
Addresses: 15.197.95.226
           75.2.122.151

> |
```

2d-2h

D

Original Message

Message ID	<6a178e407da4d_2b51cc22024@lms-canvas-worker-7b7b8996bf-x6f8m.mail>
Created at:	Thu, May 28, 2026 at 7:37 AM (Delivered after 4 seconds)
From:	"An ninh di động (2526II_INT3324_1)" <uet_announcement_03@vnu.edu.vn>
To:	23020588@vnu.edu.vn
Subject:	Grade Changed: Roll Call Attendance, An ninh di động (2526II_INT3324_1)
SPF:	PASS with IP 209.85.220.41 Learn more
DKIM:	'PASS' with domain vnu.edu.vn Learn more
DMARC:	'PASS' Learn more

[Download Original](#) [Copy to clipboard](#)

```

Delivered-To: 23020588@vnu.edu.vn
Received: by 2002:a05:7011:43c3:20b0:522:b225:ee0 with SMTP id pj3-n2csp827332mdb; Wed, 27 May 2026 17:37:24 -0700 (PDT)
X-Received: by 2002:a17:903:2bce:b0:2b0:5923:5194 with SMTP id d9443c01a7336-2beb0681ddbmr275994165ad.27.1779928644360; Wed, 27 May 2026 17:37:24 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1779928644; cv=none; d=google.com; s=arc-20240605;
b=IrjRf/1Zt33ALF8oZTF3jD5W1ZiZVxz/1VvYzCkSt/qq6Jyv9A/kxhy3V3ufof/1 I9n6k1L89iFEQG1a0ZL5mnlBjQe2QzHfrKw96uqJHeAmDzK/I3M6yHk07B838m2MLqLE

```

E

The three parts of the e-mail header are: Message Creation & Composition ; Submission to the Originating Mail Server and Intermediate Routing & Final Delivery

F

The Web Gateway IP: 123.30.233.81

ARIN

You searched for: **123.30.233.81**

Network	
Net Range	123.0.0.0 - 123.255.255.255
CIDR	123.0.0.0/8
Name	APNIC-123
Handle	NET-123-0-0-0-1
Parent	
Net Type	Allocated to APNIC
Organization	Asia Pacific Network Information Centre (APNIC)
Registration Date	2006-01-06
Last Updated	2010-07-30
Comments	This IP address range is not registered in the ARIN database. For details, refer to the APNIC Whois Database via WHOIS.APNIC.NET or http://wq.apnic.net/apnic-bin/whois.pl ** IMPORTANT NOTE: APNIC is the Regional Internet Registry for the Asia Pacific region. APNIC does not operate networks using this IP address range and is not able to investigate spam or abuse reports relating to these addresses. For more help, refer to http://www.apnic.net/apnic-info/whois_search2/abuse-and-spamming

The Internal Application Node IP: 10.42.16.78

Uma(17/02) GREY D x t i n h Week 11Week11 - Google DocsDNS Query Refused for R iOriginal MessageWhois-RWSMessageheader

whicis.arin.net/rest/p/10.42.16.78?s=10.42.16.78

T i d u uD o a nCybersecUma88GmailYouTubeMaps(1) CN IS-K68-UET...Google GeminiMy Drive - Google D...Student UETUET-LMS PortalGoogle Gemini

ARINLog in

You searched for: 10.42.16.78

Network	
Net Range	10.0.0.0 - 10.255.255.255
CIDR	10.0.0.0/8
Name	PRIVATE-ADDRESS-ABLK-RFC1918-IANA-RESERVED
Handle	NET-10-0-0-0-1
Parent	
Net Type	IANA Special Use
Organization	Internet Assigned Numbers Authority (IANA)
Registration Date	
Last Updated	2024-05-24

Comments

These addresses are in use by many millions of independently operated networks, which might be as small as a single computer connected to a home gateway, and are automatically configured in hundreds of millions of devices. They are only intended for use within a private context and traffic that needs to cross the Internet will need to use a different, unique address.

These addresses can be used by anyone without any need to coordinate with IANA or an Internet registry. The traffic from these addresses does not come from ICANN or IANA. We are not the source of activity you may see on logs or in

G,H

Uma(17/02) GREY D x t i n h Week 11Week11 - Google DocsDNS Query Refused for R iOriginal MessageMessageheader

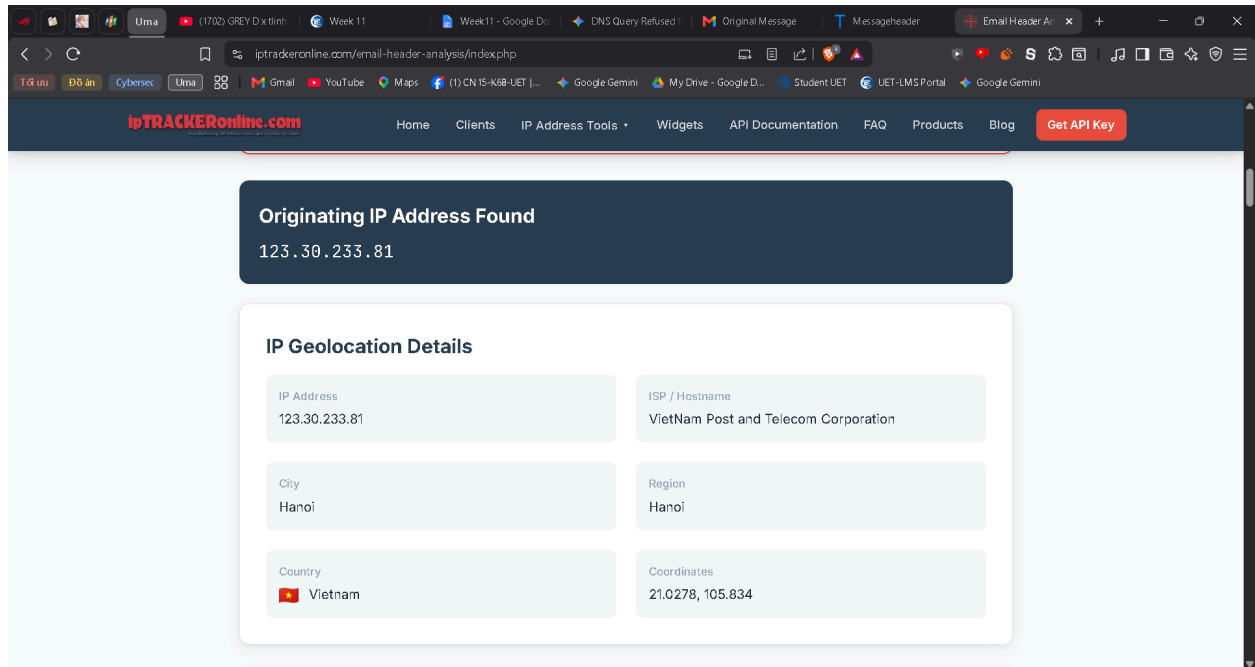
toolbox.googleapps.com/apps/messageheader/analyzeheader

T i d u uD o a nCybersecUma88GmailYouTubeMaps(1) CN IS-K68-UET...Google GeminiMy Drive - Google D...Student UETUET-LMS PortalGoogle Gemini

Messageid	6e178e407de4d_2b51cc22024@lms-canvas-worker-7b7b68996bf-x6f8m.mail				
Created at:	5/28/2026, 7:37:20 AM GMT+7 (Delivered after 4 sec)				
From:	" An ninh di động (2526LINT3324_1) " <uet_announcement_03@vnu.edu.vn>				
To:	23020988@vnu.edu.vn				
Subject:	Grade Changed: Roll Call Attendance, An ninh di động (2526LINT3324_1)				
DKIM:	pass with domain vnu.edu.vn Learn more				

#	Delay	From *	To *	Protocol	Time received
0		lms-stmp	localhost.localdomain	ESMTP	5/28/2026, 7:37:20 AM GMT+7
1	2 sec	localhost.localdomain	smtp.gmail.com		5/28/2026, 7:37:22 AM GMT+7
2	1 sec		2002:a17:902:f60b0:2ae:55eb:f82d	SMTP	5/28/2026, 7:37:23 AM GMT+7
3	1 sec	mail-sor-f41.google.com,	[Google] mx.google.com		5/28/2026, 7:37:24 AM GMT+7
4			[Google] 2002:a17:903:2bce:b0:2b0:5923:5194	SMTP	5/28/2026, 7:37:24 AM GMT+7
5			[Google] 2002:a05:7011:43c3:20b0:522:b225:ee0	SMTP	5/28/2026, 7:37:24 AM GMT+7

3b



C

This tool can exploit the coordinates and location just from an email header unlike the previous one.

17.02: SPF, DKIM, and DMARC

2a–2d

A

```
Command Prompt - nslookup x + v

Non-authoritative answer:
rit.edu text =

    "v=spf1 mx ip4:129.21.10.0/25 ip4:129.21.10.128/25 "
    "ip4:104.47.0.0/17 ip4:129.21.10.0/24 ip4:129.21.10.0/25 ip4:129.21.10.128/25 ip4:129.21.128.0/24 ip4:129.21.13.192/26 ip4:129.21.14.0/24 ip4:129.21.15.0/26 ip4:129.21.15.128/25 ip4:129.21.158.0/23 ip4:129.21.153.0/24 ip4:129.21.154.0/24 "
    "ip4:52.60.116.9 ip4:52.60.157.49 ip4:52.62.198.211 ip4:52.70.104.58 ip4:64.72.147.0/24 exists:%{i}._spf.mta.salesforce.com "
    "ip4:129.21.16.192/26 ip4:129.21.161.0/24 ip4:129.21.162.0/23 ip4:129.21.165.0/24 ip4:129.21.166.0/23 ip4:129.21.168.0/21 ip4:129.21.17.0/24 ip4:129.21.176.0/21 ip4:129.21.18.128/25 ip4:129.21.18.64/26 ip4:129.21.184.0/25 "
    "ip4:129.21.185.0/24 ip4:129.21.186.0/23 ip4:129.21.188.0/23 ip4:129.21.19.0/24 ip4:129.21.190.0/24 ip4:129.21.192.0/20 ip4:129.21.2.128/28 ip4:129.21.2.192/27 ip4:129.21.20.0/24 ip4:129.21.208.0/23 ip4:129.21.21.0/25 ip4:129.21.21.128/27 "
    "ip4:129.21.210.0/24 ip4:129.21.212.0/23 ip4:129.21.214.0/25 ip4:129.21.216.128/25 ip4:129.21.217.128/25 ip4:129.21.218.0/25 ip4:129.21.219.0/24 ip4:129.21.22.0/27 ip4:129.21.22.112/28 ip4:129.21.22.128/27 ip4:129.21.220.0/22 "
    "ip4:129.21.224.0/21 ip4:129.21.236.0/23 ip4:129.21.238.0/25 ip4:129.21.239.0/24 ip4:129.21.24.0/22 ip4:129.21.241.0/24 ip4:129.21.242.0/24 ip4:129.21.243.0/25 ip4:129.21.244.0/25 ip4:129.21.245.0/24 ip4:129.21.246.0/25 ip4:129.21.247.0/24 "
    "ip4:129.21.248.0/24 ip4:129.21.254.128/25 ip4:129.21.255.0/27 ip4:129.21.28.0/23 ip4:129.21.3.0/24 ip4:129.21.30.0/24 ip4:129.21.31.64/26 ip4:129.21.32.0/24 ip4:129.21.33.0/25 ip4:129.21.33.128/27 ip4:129.21.34.0/24 ip4:129.21.35.128/25 "
    "ip4:129.21.35.8/32 ip4:129.21.36.0/22 ip4:129.21.4.0/24 ip4:129.21.4.21/32 ip4:129.21.4.58/32 ip4:129.21.47.0/24 ip4:129.21.48.0/23 ip4:129.21.5.0/25 ip4:129.21.53.0/24 ip4:129.21.54.0/23 ip4:129.21.56.0/25 ip4:129.21.56.128/26 "
    "ip4:129.21.57.0/24 ip4:129.21.58.0/23 ip4:129.21.6.0/24 ip4:129.21.9.64/26 ip4:129.21.96.96/27 ip4:150.230.240.226 ip4:168.245.29.173 ip4:192.77.9.16/28 ip4:199.167.216.0/21 ip4:209.242.64.128/28 ip4:209.242.69.0/24 ip4:209.242.70.0/24 "
    "ip4:209.242.77.0/24 ip4:216.117.108.0/23 ip4:34.194.230.233 ip4:34.230.107.215 ip4:34.250.117.27 ip4:34.250.203.24 ip4:35.174.117.211 ip4:40.107.0.0/16 ip4:40.92.0.0/15 ip4:52.100.0.0/14 ip4:52.210.155.131 ip4:52.22.131.97 -all"
rit.edu text =

    "anthropic-domain-verification=2qymnt=MTntd4DgL6YyEaHbJUf5FKGqk"
rit.edu text =

    "hpe-greenlake-domain-verification=4b563844396f43496e68564b446f7776694a70645159306b6a534c6344355234"
rit.edu text =

    "atlassian-domain-verification=NYxaPJDKpS5S5ZcaM43zoIiTTcT16jWHscH8MZpNTXzX0LCPs7nz8YwF6RyNtDhDR"
rit.edu text =

    "openai-domain-verification=dv-aeuaxT88QLFaKvJCzy4MAoHu"
rit.edu text =

    "6C616FF21D7906B7DC2CAC02FE6155F4"
> |
```

IP Address: 129.21.4.21

Uma

(1702) GREY D - du báo thời t

Week 11

Week 11 - Google Docs

DNS Query Refused for RIT.EDU -

Whois-RWS

< > C

whcis.arin.net/rest/ip/129.21.4.21?s=12921.4.21

🔍 📄 🗑️ 🔄 🛡️

Tổ ầu Dữ ản Cybersec Uma 99

Gmail YouTube Maps (1) CN 15-K88-UET | Google Gemini My Drive - Google D... Student UET UET-LMS Portal Google Gemini

Log in

ARIN

You searched for: 129.21.4.21

Network	
Net Range	129.21.0.0 - 129.21.255.255
CIDR	129.21.0.0/16
Name	RIT
Handle	NET-129-21-0-0-1
Parent	NET129 (NET-129-0-0-0-0)
Net Type	Direct Allocation
Organization	Rochester Institute of Technology (RIT-3-Z)
Registration Date	1987-07-14
Last Updated	2022-03-16
Comments	http://www.rit.edu
RESTful Link	https://whois.arin.net/rest/net/NET-129-21-0-0-1
See Also	Related POC records
See Also	Related organization's POC records
See Also	Related delegations

IP Address: 129.21.35.8

Uma (17/02) GREY D - dự báo thời t... Week 11 Week11 - Google Docs DNS Query Refused for RIT.EDU - Whois-RWS

whcis.arin.net/rest/p/129.21.35.8?s=129.21.35.8

Enter a query above. Use * for wildcard search. By using this service, you are agreeing to the [Whois Terms of Use](#).

Search ARIN Whois with RDAP

You searched for: **129.21.35.8**

Network	
Net Range	129.21.0.0 - 129.21.255.255
CIDR	129.21.0.0/16
Name	RIT
Handle	NET-129-21-0-0-1
Parent	NET129 (NET-129-0-0-0-0)
Net Type	Direct Allocation
Organization	Rochester Institute of Technology (RIT-3-Z)
Registration Date	1987-07-14
Last Updated	2022-03-16
Comments	http://www.rit.edu
RESTful Link	https://whois.arin.net/rest/net/NET-129-21-0-0-1
See Also	Related POC records
See Also	Related organization's POC records
See Also	Related delegations

IP Address: 129.21.4.58

Uma (17/02) GREY D - dự báo thời t... Week 11 Week11 - Google Docs DNS Query Refused for RIT.EDU - Whois-RWS

whcis.arin.net/rest/p/129.21.4.58?s=129.21.4.58

Enter a query above. Use * for wildcard search. By using this service, you are agreeing to the [Whois Terms of Use](#).

Search ARIN Whois with RDAP

You searched for: **129.21.4.58**

Network	
Net Range	129.21.0.0 - 129.21.255.255
CIDR	129.21.0.0/16
Name	RIT
Handle	NET-129-21-0-0-1
Parent	NET129 (NET-129-0-0-0-0)
Net Type	Direct Allocation
Organization	Rochester Institute of Technology (RIT-3-Z)
Registration Date	1987-07-14
Last Updated	2022-03-16
Comments	http://www.rit.edu
RESTful Link	https://whois.arin.net/rest/net/NET-129-21-0-0-1
See Also	Related POC records
See Also	Related organization's POC records
See Also	Related delegations

Contact us Terms of Service Privacy Policy **ARIN** Accessibility Network Abuse Search ARIN

C

```
Command Prompt - nslookup x + v
"v=spf1 ip4:129.21.4.21/32 ip4:129.21.35.8/32 ip4:129.21.4.58/32 -all"
> chase.com
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
chase.com text =

"smartsheet-site-validation=JdBS3Kn_332V6dI9U0iq0TV3RZZXTUHL"
chase.com text =

"docusign=500adee6-4cca-451d-bcd8-2813346419c8"
chase.com text =

"google-site-verification=PfSAyrffYVUKXlc1Ew8C2IFPMkjufFSsbb0Fz_24Qt4"
chase.com text =

"google-site-verification=iZwZzo1YPL0G29U136Suzn4c1VptcA_LkvvdW0YC6B0"
chase.com text =

"wiz-domain-verification=66aa74155d5e84d10ed4b5a786a66f94063cff3b4c7e11d09fb46f027736dbf0"
chase.com text =

"atlassian-domain-verification=PZAplvJjd7scChzBMQy2d4NEwk4Bt26obCVACC7vMi0BVCoxTOV4/EB9LMexMnL"
chase.com text =

"google-site-verification=w00TwyVREI5RpqAT9hqSLZVvZcZi46578G57D1aMGeE"
chase.com text =

"sinch-domain-verification=6848bb42-da6f-49cf-8974-920af9cf1806"
chase.com text =

"atlassian-domain-verification=Ua2Fovb97Ak39kxh4koulfhVlpieV1PLhaMkdZpzINDMQGlcLV+ORgL2QmOryw+"
chase.com text =

"wiz-domain-verification=ccd3ec907fff510311f6a14b2a659fcb83adea2818bb6e78503239d2877e8657?"
chase.com text =

"onetrust-domain-verification=ccee45576c1e4fbfaa4014725a73344f"
chase.com text =
```

```
Command Prompt - nslookup x + v
chase.com text =

"wiz-domain-verification=a0d8d067bcb1cdd44255d0633a31df3ba13c82e30f27c080519b0b85ba734d32"
> set q=a
> spf.jpmchase.com
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
Name: spf.jpmchase.com
Addresses: 146.143.8.28
146.143.72.27
146.143.200.27
146.143.200.30
146.143.8.18
146.143.136.28
146.143.8.27
146.143.200.18
146.143.200.22
146.143.136.25
146.143.136.18
146.143.8.23
146.143.200.24
146.143.136.20
146.143.136.24
146.143.8.25
146.143.72.17
146.143.136.27
146.143.200.19
146.143.136.21
146.143.8.22
146.143.72.24
146.143.72.26
146.143.72.29
146.143.200.26
146.143.8.30
146.143.200.21
146.143.136.29
146.143.72.28
146.143.72.19
146.143.136.19
```

```
Command Prompt - nslookup  X + v
146.143.200.21
146.143.136.29
146.143.72.28
146.143.72.19
146.143.136.19
146.143.72.25
146.143.8.29
146.143.136.30
146.143.8.26
146.143.72.21
146.143.200.28
146.143.200.29
146.143.200.20
146.143.136.23
146.143.136.17
146.143.8.24
146.143.136.22
146.143.72.23
146.143.200.17
146.143.8.17
146.143.8.20
146.143.8.19
146.143.200.25
146.143.200.23
146.143.72.18
146.143.72.22
146.143.136.26
146.143.72.20
146.143.72.30
146.143.8.21

> set q=txt
> tpo.chase.com
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
tpo.chase.com text =

"v=spf1 ip4:68.233.76.14/32 ip4:63.150.74.35/32 ip4:198.64.159.0/24 ip4:198.104.137.206/32 ip4:161.58.88.0/24 -all"
> |
```

```
Command Prompt - nslookup  X + v
> tpo.chase.com
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
tpo.chase.com text =

"v=spf1 ip4:68.233.76.14/32 ip4:63.150.74.35/32 ip4:198.64.159.0/24 ip4:198.104.137.206/32 ip4:161.58.88.0/24 -all"
> syr.edu
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
syr.edu text =

"v=spf1 mx:spf.syr.edu include:spf.protection.outlook.com -all"
syr.edu text =

"anthropic-domain-verification-bbywt6=lAujtISTVyzitNixCeNUfyQOI"
syr.edu text =

"jamf-site-verification=M5etiTqGdkTkdsFNzn7Yw"
syr.edu text =

"38ful0fo2v6cjntnk0jdaaovaf"
syr.edu text =

"airtable-verification=75fc4c60b061a07e5a1973a39c2c6032"
syr.edu text =

"d365mktkey=xtwwJrx4xyMxzKlKtGfXrtAD3RrKPssxi4xghrW9LCYx"
syr.edu text =

"atlassian-sending-domain-verification=e48074eb-dc35-4c0a-801d-15b7e7152ce8"
syr.edu text =

"d365mktkey=k3qWbfxG7YqGLgbdErhW3jrkxJVAdkRCmRAMQ9e0zB4x"
syr.edu text =

"atlassian-domain-verification=YmuP0w5kKYcGbbj2l0Hznj1ktloqUcXNbW2ArTDpxPJfUz/znB/dhnAJa6cu17QQ"
> |
```

```
Command Prompt - nslookup  X + v
Non-authoritative answer:
syr.edu text =
"v=spf1 mx:spf.syr.edu include:spf.protection.outlook.com -all"
syr.edu text =
"anthropic-domain-verification-bbywt6=IAujtISTVyzitNIXCeNUfyQOI"
syr.edu text =
"jamf-site-verification=M5etiTqGdkTkd5sFNzn7Yw"
syr.edu text =
"38fu1ofo2v6cjntnk0jdaaovaf"
syr.edu text =
"airtable-verification=75fc4c60b061a07e5a1973a39c2c6032"
syr.edu text =
"d365mktkey=xtwwJrx4xyMxzKlKtGfXrtAD3RrKPsxxi4xghrW9LCYx"
syr.edu text =
"atlassian-sending-domain-verification=e48074eb-dc35-4c0a-801d-15b7e7152ce8"
syr.edu text =
"d365mktkey=k3qWbfxG7YqGLgbdErhW3jrkRJVAdkRCmRAMQ9e0zB4x"
syr.edu text =
"atlassian-domain-verification=YmuP0w5kKYcGbbj2l0Hznj1ktloqUcXNbW2ArTDpxPjFUr/znB/dhnAJa6cu17QQ"
> set q=mx
> spf.syr.edu
Server: [8.8.8.8]
Address: 8.8.8.8
Non-authoritative answer:
spf.syr.edu MX preference = 10, mail exchanger = exchange.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = mx-ext.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = mx-int.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = smtp-relay.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = syr-edu.mail.protection.outlook.com
>
```

D

My “spf=pass” in one of my e-mails from the previous lab exercise in Gmail:
spf=pass (google.com: domain of uet_announcement_03@vnu.edu.vn designates
209.85.220.41 as permitted sender) smtp.mailfrom=uet_announcement_03@vnu.edu.vn;

3a-3b

A

dkim=pass header.i=@vnu.edu.vn header.s=google header.b=ONXSkEnz;

DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=vnu.edu.vn; s=google;
t=1779928644; x=1780533444; darn=vnu.edu.vn;
h=auto-submitted:content-transfer-encoding:mime-version:subject
:message-id:to:reply-to:date:from:from:to:cc:subject:date:message-id :reply-to;
bh=ZgArletEhRq3PYgsHT5HXfuLW33337ZRofHasfRtBt0=;
b=ONXSkEnzBa/YsLNQ+geLdZg8T8E9fh3CP2MTpsXia/axHQ+ZNICN9K8/FFEbXk
GfEC...

B

```
Command Prompt - nslookup  X + v

"jamf-site-verification=M5etiTqGdkTkdsFNzn7Yw"
syr.edu text =

"38fulof02v6cjntnk0jdaaovaf"
syr.edu text =

"airtable-verification=75fc4c60b061a07e5a1973a39c2c6032"
syr.edu text =

"d365mktkey=xtwwJrx4xyMxzKLtGfYrtAD3RrKPssxi4xghrW9LCYx"
syr.edu text =

"atlassian-sending-domain-verification=e48074eb-dc35-4c0a-801d-15b7e7152ce8"
syr.edu text =

"d365mktkey=k3qWbfXG7YqGLgbdErhW3jrkRJVAdkRCmRAMQ9e0zB4x"
syr.edu text =

"atlassian-domain-verification=YmuP0w5kKYcGbbj2l0Hznj1ktloqUcXNbW2ArTDpxPjFUr/znB/dhnAJa6cu17QQ"
> set q=mx
> spf.syr.edu
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
spf.syr.edu MX preference = 10, mail exchanger = exchange.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = mx-ext.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = mx-int.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = smtp-relay.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = syr-edu.mail.protection.outlook.com
> set q=txt
> rit1608._domainkey.rit.edu.
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
rit1608._domainkey.rit.edu text =

"v=DKIM1; p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDXq97I7+PXPn39GE0fTg3Ypn3GmtrA2LsQCNj+q2iyK0CewalH9RiM7bF7SbrDFzFF2iQv1PPcHfBpBihpnnnPxrStxK+UZDEb
5b9VDyUr+MCWsk7L4w8Fr6DA8Bwn4X5Ykzp6rvJe0AFoFg8d/0dhxQyA/1Rg23aXE6ee6S5QjFQIDAQAB;"
> |
```

4a-4b

A

```
Command Prompt - nslookup  X + v

"d365mktkey=k3qWbfXG7YqGLgbdErhW3jrkRJVAdkRCmRAMQ9e0zB4x"
syr.edu text =

"atlassian-domain-verification=YmuP0w5kKYcGbbj2l0Hznj1ktloqUcXNbW2ArTDpxPjFUr/znB/dhnAJa6cu17QQ"
> set q=mx
> spf.syr.edu
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
spf.syr.edu MX preference = 10, mail exchanger = exchange.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = mx-ext.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = mx-int.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = smtp-relay.syr.edu
spf.syr.edu MX preference = 10, mail exchanger = syr-edu.mail.protection.outlook.com
> set q=txt
> rit1608._domainkey.rit.edu.
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
rit1608._domainkey.rit.edu text =

"v=DKIM1; p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDXq97I7+PXPn39GE0fTg3Ypn3GmtrA2LsQCNj+q2iyK0CewalH9RiM7bF7SbrDFzFF2iQv1PPcHfBpBihpnnnPxrStxK+UZDEb
5b9VDyUr+MCWsk7L4w8Fr6DA8Bwn4X5Ykzp6rvJe0AFoFg8d/0dhxQyA/1Rg23aXE6ee6S5QjFQIDAQAB;"
> set q=txt
> dmarc.paypal.com.
Server: [8.8.8.8]
Address: 8.8.8.8

*** [8.8.8.8] can't find dmarc.paypal.com.: Non-existent domain
> _dmarc.paypal.com.
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
_dmarc.paypal.com text =

"v=DMARC1; p=reject; rua=mailto:d@rua.agari.com; ruf=mailto:d@ruf.agari.com"
> |
```

```
Command Prompt - nslookup
Address: 8.8.8.8

Non-authoritative answer:
spf.syr.edu      MX preference = 10, mail exchanger = exchange.syr.edu
spf.syr.edu      MX preference = 10, mail exchanger = mx-ext.syr.edu
spf.syr.edu      MX preference = 10, mail exchanger = mx-int.syr.edu
spf.syr.edu      MX preference = 10, mail exchanger = smtp-relay.syr.edu
spf.syr.edu      MX preference = 10, mail exchanger = syr-edu.mail.protection.outlook.com
> set q=txt
> rit1608._domainkey.rit.edu.
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
rit1608._domainkey.rit.edu      text =

"v=DKIM1; p=MIGfMA0GCSqGSIb3DQEBQUAA4GNADCBiQKBgQDXq97I7+PXPn39GE0fTg3Ypn3GmtrA2LsQCnj+q2iyKOCewa1H9RiM7bF7SbrDFzFF2iQv1PPcHfBpBihpnnnPxr5txK+UZDEb
5b9VDyUr+MCWsk7L4w8Fr6DA8Bwn4X5Ykzp6rvJe0AFofG8d/0dhxQyA/1Rg23aXE6ee65QjFQIDAQAB;"
> set q=txt
> dmarc.paypal.com.
Server: [8.8.8.8]
Address: 8.8.8.8

*** [8.8.8.8] can't find dmarc.paypal.com.: Non-existent domain
> _dmarc.paypal.com.
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
_dmarc.paypal.com      text =

"v=DMARC1; p=reject; rua=mailto:d@rua.agari.com; ruf=mailto:d@ruf.agari.com"
> _dmarc.linkedin.com.
Server: [8.8.8.8]
Address: 8.8.8.8

Non-authoritative answer:
_dmarc.linkedin.com      text =

"v=DMARC1; p=reject; rua=mailto:d@rua.agari.com,mailto:yfy3q-9359@rua.dmarc.emailanalyst.com; ruf=mailto:d@ruf.agari.com"
> |
```

B
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=vnu.edu.vn;

Lab Analysis

1. What is an SMTP server?

An **SMTP server** is an application that handles the transmission, routing, and delivery of email messages across the network using the Simple Mail Transfer Protocol.

2. How is an SMTP server both an MTA client and an MTA server?

It changes roles depending on the direction of the email:

- **MTA Server:** It acts as a server when **receiving** incoming emails from a user or another mail system.
- **MTA Client:** It acts as a client when **sending** or relaying those emails out to an external destination mail system.

3. In which direction are e-mail headers read, and why?

Emails are logged chronologically. The message is created at the bottom, and each mail server it passes through stamps its **Received:** tracking info on **top** of the previous record.

Key Term Quiz

Use the terms from the list to complete the sentences that follow. DKIM DMARC SPF

1. Receiving mail servers can ignore **SPF** records.
2. The use of a public key by **DKIM** uses a public key.
3. The usage of **DMARC** depends on the other two being configured.